

Sleep as Attractor Basin Navigation Cost: A Unified Geometric Theory of Sleep Function, Topology, and Deep-Time Survival

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Manuscript deposited as reasoning artifact series, OrganismCore, 2026-03-18.

Repository: github.com/Eric-Robert-Lawson/attractor-oncology

Theory paper: [doi:10.5281/zenodo.19094935](https://doi.org/10.5281/zenodo.19094935)

Abstract

We present a unified geometric theory of sleep derived from the attractor landscape framework $S + N + G \rightarrow R$, where S is the structural substrate (genome and somatic integrity), N is the coherence gradient field (circadian clock, social environment, and light), and G is the global attractor landscape, physically implemented as the organism's epigenetic state. The framework makes three inseparable predictions about rest: (1) *Quantity* — the total cost of occupying a position in the attractor landscape must be discharged per cycle, and duration of discharge tracks total cost; (2) *Topology* — the architecture of the discharge process is itself a geometric prediction of the landscape position, not an independent biological variable; and (3) *Substrate* — the cost of basin navigation is fundamentally oxidative, with reactive oxygen species (ROS) accumulation as the proximate death mechanism in sleep deprivation (Vaccaro et al., 2020), and the glymphatic system as the topological mechanism of discharge within consolidated bilateral sleep.

From these three predictions we derive: eight sleep topologies (T1 through T8); two deep-time survival strategies (Lock and Navigator); an unstable zone; a complete three-channel sonar instrument (duration, topology, behavioural flexibility); and the dimensional invariant — the oldest surviving lineages sleep the least and the minimum of the sleep spectrum is the maximum of the survival spectrum.

We further identify that (a) the four dominant sleep function theories in the current literature (synaptic homeostasis, energy conservation, glymphatic clearance, memory consolidation) are correct descriptions of mechanisms within a single topology class (T3) and are subsumed by the framework as special cases; (b) the derivation chain from cancer biology (FOXA1/EZH2 ratio; Lawson, 2024) closes on itself through BMAL1 — loss of the master circadian clock gene makes the cellular repair basin epigenetically inaccessible, producing a cancer attractor at the cellular level geometrically identical to the organism-level failure it initiated; (c) social information is a primary N field co-equal to light, with dominance rank negatively correlating with sleep quality proportional to the social navigational demand of the dominant position; and (d) artificial light at night (ALAN) is a global primary N field disruption operating simultaneously across thousands of species.

Eleven a-priori predictions (P-INVARIANT-1 through P-INVARIANT-11) and five closure predictions (P-CLOSURE-1 through P-CLOSURE-5) are stated. None are derivable from any existing sleep function theory. All were stated before literature confirmation was sought.

1. Introduction: The Problem With Four Theories

Sleep is universal across animal life and has persisted across 600 million years of evolution despite imposing an obvious and severe fitness cost: a sleeping organism cannot feed, reproduce, or detect predators. The

persistence of sleep therefore implies a benefit sufficiently large to outweigh this cost continuously across deep time.

The current literature offers four competing frameworks for what that benefit is:

1. **Synaptic Homeostasis (SHY):** Sleep downscale synaptic potentiation accumulated during waking, preventing synaptic saturation and restoring the brain's capacity for new learning [?].
2. **Energy conservation:** Sleep reduces metabolic demand and allows cellular energy reserve replenishment [?].
3. **Glymphatic clearance:** Sleep drives cerebrospinal fluid pulsation through perivascular channels, clearing neurotoxic metabolites including amyloid-beta [?].
4. **Memory consolidation:** Sleep reactivates and stabilises memory traces formed during waking [?].

The 2023–2024 consensus is explicit: no single theory explains all aspects of sleep [?]. The frameworks are treated as complementary and overlapping. This theoretical pluralism is the symptom of a missing geometric structure.

The problem: None of these four frameworks generates the following observations:

- The oldest surviving lineages on Earth sleep the least.
- Organisms that sleep almost nothing (horseshoe crabs, sponges, nautilus) have survived every mass extinction for hundreds of millions of years.
- Organisms that sleep the most (bats, koalas, cats) are not the most cognitively complex.
- The dolphin has dissolved the barrier between waking and sleep and permanently occupies an intermediate state.
- The octopus has independently evolved a two-phase sleep architecture in a brain with no cerebral cortex.
- A crocodilian sleeps 2–6 hours of true bilateral EEG sleep and 12–16 hours of distributed rest, not because it is cognitively simple but because its ectothermic metabolic architecture determines the only topologically feasible discharge architecture for its navigational demand.

These observations are not anomalies. They are the geometric signal. The theory that explains them must be geometric, not mechanistic.

We propose such a theory.

2. The Attractor Geometry Framework

2.1. The Fundamental Equation

$$S + N + G \rightarrow R$$

- S = structural substrate (genome, epigenetically accessible gene programmes, somatic tissue integrity, stem cell reserve)
- N = coherence gradient field (light, social information, and circadian clock genes as the molecular implementation of N at the cellular level)
- G = global attractor landscape, physically implemented as the organism's epigenetic state: histone modification patterns, DNA methylation, and chromatin accessibility across the genome
- R = resulting state or phenotype

An organism's position in G determines four properties simultaneously: (1) the total cost of occupying that position; (2) the survival trajectory; (3) the topology of the discharge process; and (4) the biochemical substrate of the cost.

2.2. Sleep as Basin-Navigation Cost Discharge

We define sleep as the generic process by which the accumulated cost of basin occupation in G is discharged. The cost has three axes:

Axis 1 — Between-basin switching cost. The organism navigates between multiple accessible basins in G . Each transition requires bilateral synaptic reconfiguration and generates ROS in proportion to the neural firing intensity of the transition.

Axis 2 — Within-basin metabolic cost. A single high-cost basin consumes all available metabolic resources. Sleep is required for metabolic recovery and oxidative debt clearance rather than synaptic reconfiguration.

Axis 3 — Within-basin computational intensity. Extreme neural computation within a single specialised basin generates high ROS from neural firing intensity. Discharge is required for both synaptic and oxidative cost.

2.3. The Oxidative Substrate

The cost of navigation is fundamentally oxidative. This is confirmed by the finding that sleep deprivation kills via reactive oxygen species accumulation specifically in the intestinal epithelium of *Drosophila*, and that antioxidant supplementation prevents death under total sleep deprivation even without additional sleep [?]. The gut is the tissue most directly responsible for substrate acquisition driving the navigation, and its epithelium bears the highest peripheral oxidative cost.

Sleep discharges two costs simultaneously: the synaptic cost (via slow-wave downscaling, the Synaptic Homeostasis mechanism [?]) and the oxidative cost (via glymphatic pulsation during NREM slow-wave sleep, driven by interstitial space expansion and AQP4-mediated CSF flow through perivascular channels [?]). The glymphatic system is the physical mechanism of oxidative cost discharge within the consolidated bilateral sleep topology.

2.4. G is the Epigenetic State

The attractor landscape G is not an abstract theoretical object. Its physical substrate is the epigenetic state of the organism's cells.

Basin depths in G correspond to the depth of epigenetic entrenchment of the corresponding state. Barrier heights correspond to the chromatin reconfiguration energy required to transition between states. Sleep maintains G daily through sleep-dependent histone deacetylation (restoring chromatin accessibility patterns to their functional state) and DNA methylation maintenance [?].

Chronic sleep deprivation degrades G irreversibly: progressive histone modification dysregulation, DNA methylation drift, pleiotrophin depletion leading to neuronal death (removing basins permanently), and NF- κ B-driven inflammatory landscape remodelling [?]. The irreversible components of chronic sleep deprivation damage are G degradation past the restoration threshold.

3. The Eight Sleep Topologies

The framework predicts that different landscape positions produce different sleep architectures. Each topology is the only feasible discharge architecture for its corresponding landscape position given the full constraint set.

3.1. T1: Metabolic Quiescence

Deepest Lock organisms with minimal neural architecture. Zero navigational demand. Zero synaptic reconfiguration cost. Minimal ROS generation. No sleep in any neurological sense; circadian biochemical rhythms only.

Confirmed: Sponge (~600 Myr), horseshoe crab (~450 Myr), most jellyfish (500–700 Myr).

3.2. T2: Distributed Bilateral Rest

Ectothermic Navigators. Navigator-level navigational cost but ROS generation rate an order of magnitude lower than endothermy. Energy budget for full bilateral wakefulness limited per day.

True bilateral EEG sleep for 2–6h discharging synaptic cost. Extended distributed bilateral rest for 12–16h clearing the low-rate oxidative debt continuously. Total Navigator-equivalent discharge. Why bilateral and not unihemispheric: the switching cost is between ecological basins, bilaterally symmetric, requiring no hemispheric asymmetry in discharge.

Confirmed: Crocodilian (bilateral EEG throughout rest periods, Kelly & Lesku, 2015, 2019). One-eye-open vigilance is behavioural (brainstem alerting), not unihemispheric in the EEG sense.

3.3. T3: Consolidated Bilateral Deep Sleep

Endothermic organisms. High ROS generation rate during waking requires concentrated high-efficiency glymphatic clearance during sustained NREM. Bilateral synaptic reconfiguration discharged during same concentrated window.

Confirmed: All endothermic organisms. Duration within T3 is a valid direct proxy for total cost because metabolic architecture is held constant.

3.4. T4: Permanent Basin Omega — Dissolved Barrier

Open-ocean Navigators where full bilateral sleep is lethal. The cost of entering the sleep basin's exit barrier exceeds the accumulated cost of remaining partially awake indefinitely. Barrier between waking and sleep is dissolved. Hemispheric alternation in continuous rotation.

Confirmed: Dolphin, porpoise, some seals. Frigatebirds during multi-day oceanic flight.

3.5. T5: Fragmented Shallow Sleep — External Suppression

Deep single basin under chronic external pressure interrupting sleep before discharge completes. Sleep basin is behaviourally shallow, not geometrically shallow. Debt accumulates and is discharged as rebound when pressure is removed. Rebound is the definitive diagnostic.

Confirmed: Giraffe.

3.6. T6: Transition-Cost Sleep — Developmental Basin Crossing

Sleep required not to discharge ongoing occupation cost but to pay the cost of a developmental basin transition. Duration proportional to transition magnitude, not to ongoing navigational demand. Cannot be postponed without developmental failure.

Confirmed: *C. elegans* lethargus at each larval transition [?]. Broadly, all animals with discrete developmental basin transitions.

3.7. T7: Immune-Basin Sleep — Defensive Basin Occupation

Pathogen invasion creates a new high-cost basin: the immune response. Sleep is the discharge process for immune-basin occupation cost. Cytokines are the N field signal specifying immune basin entry. Duration proportional to immune response intensity.

Confirmed: *C. elegans* sickness quiescence [?]. Human and mammalian fever sleep. Confirmed across all major animal phyla.

3.8. T8: Suspended State with Blocked Discharge

Hibernation/torpor. Both navigational activity and sleep architecture are suspended by temperature constraints. Homeostatic pressure continues to build in the complete absence of navigational activity. Periodic arousals driven by homeostatic pressure reaching threshold — metabolically expensive warming purely to enable intense slow-wave sleep discharge. Post-emergence rebound qualitatively different from T5: maximum debt with zero navigational cause.

Confirmed: Ground squirrel, bear, bat, hedgehog. Post-emergence rebound sleep proportional to hibernation duration.

3.9. The Eight-Topology Prediction

The eight topologies are not a closed taxonomy. They are the geometrically derivable solutions given the known constraint set. Any new topology discovered empirically will be derivable from the same principles: the interaction of navigational cost structure, ROS generation rate, metabolic architecture, environmental lethality constraints on full sleep, developmental transition requirements, and immune-basin occupation. The framework is not a list. It is a geometry.

4. The Two Survival Strategies and the Unstable Zone

4.1. The Lock Strategy

Single basin. Extremely deep. Very high exit barrier. Minimal metabolism. Minimal ROS generation. Sleep duration approaching zero in simplest forms. Topology T1 or T2. Survival horizon: 300–600 Myr+.

Convergent across seven independent phyla: Porifera, Cnidaria, Mollusca, Chelicerata, Chondrichthyes, Sarcopterygii, Lepidosauria.

Survival mechanism: basin so deep it is perturbation-invariant. The Permian–Triassic event eliminated 96% of marine species. The horseshoe crab (T1, near-zero sleep, 450 Myr) survived.

4.2. The Navigator Strategy

Multiple basins. Shallow barriers. High switching capacity. Ecological generalism. Moderate to high sleep equivalent. Topology T2, T3, or T4 depending on metabolic architecture and environmental constraints. Survival horizon: 100–300 Myr.

Convergent across six+ independent lineages across T2, T3, and T4. The critical confirmation: octopus (Cephalopoda, ~300 Myr) has independently evolved a two-phase sleep (quiet plus active REM-like) in a brain with no cerebral cortex [?]. The same topology from completely different neural architecture. Topology is determined by navigational demand, not by neural architecture.

4.3. The Unstable Zone

Shallow single basin. High barriers to alternatives. Not deep enough for perturbation-invariance. Not flexible enough to reach alternatives when niche disappears. Extinction concentrated here. Examples: *T. rex*, mammoth, dodo.

4.4. The Dimensional Invariant

The oldest surviving lineages sleep the least. The minimum of the sleep spectrum is the maximum of the survival spectrum. This is not a coincidence. This is the geometry.

The survival–sleep duration relationship is U-shaped with a flat minimum. The minimum of sleep duration corresponds to the maximum survival horizon (Lock strategy). The moderate range is the Navigator survival zone. The high-duration range is uncertain — survival depends on niche stability or Basin Ω depth.

5. The N Field: Clock, Social Information, and Light

The coherence gradient field N has three confirmed physical implementations:

5.1. The Circadian Molecular Clock

BMAL1/CLOCK/PER/CRY forms the molecular N field that specifies when cells and organisms transition between active and rest/repair basins. BMAL1/CLOCK acts as a histone acetyltransferase — it directly opens chromatin at repair gene loci, implementing the transition into the repair basin at the molecular level.

BMAL1 loss at the cellular level is the molecular equivalent of an organism that cannot enter its sleep basin. BMAL1 acts as a tumour suppressor [?]: its loss makes the repair/differentiation basin epigenetically inaccessible and stabilises the cancer attractor.

This closes the derivation chain. The framework began with a cancer cell locked in the cancer attractor (FOXA1/EZH2 ratio, Lawson 2024). It ends by identifying that BMAL1 loss is the molecular mechanism by which the repair basin becomes inaccessible — the same geometric failure. Cancer is cellular sleep deprivation. The chain is circular.

5.2. Social Information as Primary Zeitgeber

Social cues are confirmed co-primary zeitgebers, entraining the circadian clock independently of light [?]. The social N field synchronises group sleep basins into a coupled collective attractor.

Wild chacma baboons: dominant individuals experience less and lower-quality sleep than subordinates [?]. Dominant baboons sleep closer to more group members, experience more social N field signals during the night, and cannot fully enter the sleep basin without losing the social landscape position that dominance requires. Dominance is a landscape position whose navigational cost does not cease at night.

5.3. Light and Artificial Light at Night

Light is the primary physical zeitgeber. ALAN is expanding at 2.2% per year globally [?], disrupting the primary N field for all phototactic and circadian-entrained species simultaneously. 600+ studies document consequences across birds, insects, marine organisms, mammals, and plants.

ALAN is not a minor ecological inconvenience. It is a global disruption of the primary N field that specifies sleep basin entry for all life. The consequences are geometric: organisms whose N field is corrupted lose reliable access to the basins the field was specifying.

6. Subsuming the Four Existing Theories

6.1. Why All Four Are Correct and Incomplete

The four dominant theories are correct descriptions of mechanisms within T3 (consolidated bilateral deep sleep in endothermic organisms). This topology covers the majority of the published comparative sleep literature because most studied organisms are endothermic vertebrates.

SHY describes the synaptic component of Axis 1 and Axis 3 cost discharge within T3.

Energy conservation describes the consequence of reduced navigational activity during Axis 2 cost recovery within T3.

Glymphatic clearance describes the physical mechanism by which the oxidative cost substrate is cleared during T3 via NREM-driven CSF pulsation.

Memory consolidation describes a downstream consequence of selective synaptic downscaling within T3.

None generates the dimensional invariant. None predicts that the oldest surviving lineages sleep the least. None predicts convergent topologies across independent phylogenies. None predicts that sleep topology is a geometric property of landscape position. The dimensional invariant contains all four as projections of a single geometric structure.

6.2. What the Framework Adds

The framework generates eleven a-priori predictions (P-INVARIANT-1 through P-INVARIANT-11, stated in the Dimensional Invariant v3.0 reasoning artifact, 2026-03-18) and five closure predictions (P-CLOSURE-1 through P-CLOSURE-5, stated in the Derivation Chain Closure reasoning artifact, 2026-03-18), none of which are derivable from any of the four existing theories, including:

- Sleep duration will negatively correlate with lineage age across all phyla, topology-calibrated.
- Topology 2 will be found specifically and exclusively in ectothermic Navigators.

- T6 quiescence duration will scale with basin transition magnitude, not ongoing navigational demand.
- T8 post-emergence rebound will be predicted by hibernation duration alone.
- BMAL1 restoration in cancer cells will partially restore differentiation basin chromatin accessibility.
- ALAN exposure will produce measurable G degradation proportional to duration and intensity.

7. Convergence Across Independent Phyla

7.1. The Lock Converges Across Seven Independent Phyla

Seven independent phyla have arrived at the same geometric strategy — deep single basin, minimal metabolism, minimal sleep — across evolutionary timescales of 200–600 Myr. The convergence is not taxonomic. It is geometric: the Lock strategy is the deepest available position in G for organisms whose ecological niche is stable and non-navigational.

The transition from T1 to T2 within the Lock as neural complexity increases is itself a geometric prediction: added neural architecture adds synaptic cost to discharge and ROS from neural firing, but the basin remains deep enough to keep total cost minimal. Topology shifts but duration remains in the Lock range.

7.2. The Navigator Converges Across Multiple Topologies

The Navigator strategy has independently converged across T2, T3, and T4 in at least seven lineages. The strategy is the same — ecological generalism as Basin Ω of ecological flexibility — expressed through the only topologically feasible discharge architecture given each lineage's full constraint set.

The octopus confirmation is the strongest single convergence in the dataset: 500 Myr of separate evolution, completely different neural architecture (no cerebral cortex, distributed ganglionic organisation), confirmed Navigator ecological strategy, and independently evolved two-phase sleep (quiet plus active REM-like) [?]. Topology is determined by navigational demand. Biology is not.

7.3. The Unstable Zone Does Not Converge

No convergent feature exists in the unstable zone except instability. This non-convergence confirms the framework: there is no deep basin in the unstable zone to converge on. Extinction is the non-convergent attractor.

8. The Three-Channel Sonar Instrument

The complete sonar instrument for landscape position determination has three primary channels and two supplementary diagnostics:

Channel 1 — Duration: Measures total discharge cost. Valid proxy within a single topology class. Ambiguous across topology classes.

Channel 2 — Topology: Identifies which of the eight discharge architectures the organism uses. Removes topology-class ambiguity from duration. A crocodilian (T2, 2–6h true sleep) and an elephant (T3, 2h true sleep) have similar true sleep durations but completely different cost structures. Topology disambiguates.

Channel 3 — Behavioural flexibility: Discriminates Navigator from Unstable Zone when both occupy T3. Discriminates T5 (flexibility impaired under pressure) from basin restructuring (flexibility maintained within new basin).

Supplementary 1 — Rebound response: T5 diagnostic. T8 diagnostic (distinct from T5: debt-to-activity ratio is infinite in T8). Not required once topology is identified but required when topology data are unavailable.

Supplementary 2 — ROS rate: Predicted to discriminate ectothermic T2 from endothermic T3 at equivalent navigational cost levels. Testable via metabolic rate measurement during active period.

The migratory white-crowned sparrow provides the critical validation of the sonar instrument: minimal sleep during migration with no rebound is basin restructuring (lower actual demand in the migratory attractor), not T5 debt suppression. The performance diagnostic — maintained within-basin performance, not impaired — confirms this. Duration alone would misclassify the sparrow as approaching Lock range. The full three-channel sonar correctly identifies basin restructuring.

9. Discussion: The Scale Invariance of the Framework

9.1. From Bacterium to Biosphere

The framework operates identically across eleven orders of magnitude of biological scale:

Single-celled organisms Bacterial persister cells demonstrate Lock/Navigator portfolio strategy at sub-organismal population level. Specialist bacteria show Lock-equivalent metabolic quiescence. Generalist bacteria show moderate distributed rest. The dimensional invariant is detectable below the level of the nervous system.

Developmental organisms T6 topology: *C. elegans* lethargus is mandatory sleep at developmental basin transitions. Adolescent synaptic pruning is T6 operating as a prolonged landscape construction process — SWA declining across adolescence is the geometric signature of a landscape approaching completion.

Individual organisms The five primary sleep functions (discharge of Axes 1, 2, and 3 costs, somatic G maintenance via growth hormone-driven stem cell renewal, and G restoration via epigenetic maintenance) are all expressed within the same sleep window.

Social groups Social information is a primary N field. Group sleep is a coupled attractor. Dominance rank imposes navigational cost that does not cease at night. The baboon hierarchy is a landscape whose navigational demands are expressed in sleep quality.

Species across deep time The dimensional invariant. The oldest surviving lineages sleep the least. Seven Lock phyla. Seven Navigator lineages across three topologies. The unstable zone does not converge. The geometry is consistent across 600 Myr.

The biosphere ALAN is a global primary N field disruption affecting all species simultaneously. It is remodelling the N field component of the $S + N + G \rightarrow R$ equation for thousands of species at once, with measurable eco-evolutionary consequences already underway.

9.2. What Remains Open

The framework does not yet predict the topological consequences of a basin that deepens in real time at inter-generational scale (the human Basin B*). It does not specify the threshold at which G degradation becomes irreversible for any given organism. It does not model the dynamics of the ALAN eco-evolutionary feedback — whether species will adapt their S substrate to function with a corrupted N field before the disruption causes population collapse.

These are the honest limits of the current derivation.

9.3. The Framework is Not Reductive

The existing four sleep function theories are not wrong. They are not superseded. They are confirmed as correct descriptions of mechanisms within their scope (T3, endothermic organisms). The framework does not replace them. It provides the geometric structure that explains why they are all correct, why none of them alone is sufficient, and what they are all projections of.

The same relationship holds at every scale. The molecular biology of BMAL1 is not superseded by the geometric framework. It is the physical implementation of the N field at the cellular level. The glymphatic anatomy is not superseded. It is the physical implementation of T3 oxidative discharge. The Waddington epigenetic landscape is not superseded. It is G .

The geometry is the structure. The mechanisms are the geometry made physical.

10. Conclusion

We have presented a unified geometric theory of sleep derived from the attractor landscape framework $S + N + G \rightarrow R$, in which:

- G is physically the epigenetic state
- N is physically the circadian clock, social information, and light — each a co-equal zeitgeber
- S is the structural substrate maintained by sleep-dependent growth hormone release
- Sleep is the process that simultaneously maintains all three components within a single consolidated window

The framework makes the dimensional invariant: across all life, at all scales, across all of deep time, the oldest surviving lineages sleep the least. The minimum of the sleep spectrum is the maximum of the survival spectrum.

The framework identifies eight sleep topologies as geometric predictions of landscape position, not empirical categories.

The framework closes the derivation chain: BMAL1 loss produces a cancer attractor by making the repair basin epigenetically inaccessible. Cancer is cellular sleep deprivation. The geometry is the same object described from the molecular to the civilisational scale.

Sleep is not a behaviour. Sleep is the daily maintenance of the landscape that makes all behaviour possible. Survival is the depth of that landscape.

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Manuscript ID: ATTRACTOR_GEOMETRY_SLEEP_THEORY_PAPER_v1.0

Date: 2026-03-18

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Status: Theory paper. Self-contained. Cites and supersedes the reasoning artifact series as the primary theoretical statement.

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